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# =====
# EPPS6356 - Data Visualization Assignment 4
# Happy Planet Index
# =====

# =====
# Chart 1: Variable-width Column Chart – HPI by Continent
# =====

library(readxl)
library(dplyr)
library(ggplot2)
library(scales)

source("shared_theme.R")

# ---- Load data ----
setwd("D:/PhD Study/Fall 2026/EPPS 6356")
path <- "Happy-Planet-Index-2006-2025-public-data-set.xlsx"

hpi_raw <- read_excel(path, sheet = "All Data", skip = 1)
hpi_clean <- hpi_raw %>%
  filter(!is.na(HPI))

continent_lookup <- tibble(
  Continent = 1:8,
  Continent_name = c("Latin America", "N America & Oceania", "Western Europe",
    "Middle East & N. Africa", "Africa", "South Asia",
    "Eastern Europe & Central Asia", "East Asia")
)

# ---- Aggregate: one population snapshot per country, mean HPI across years
----
pop_snapshot <- hpi_clean %>%
  group_by(Country) %>%
  slice_max(Year, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  left_join(continent_lookup, by = "Continent") %>%
  group_by(Continent_name) %>%
  summarize(total_pop = sum(Population, na.rm = TRUE))

chart1_data <- hpi_clean %>%
  left_join(continent_lookup, by = "Continent") %>%
  group_by(Continent_name) %>%
  summarize(mean_hpi = mean(HPI, na.rm = TRUE)) %>%
  left_join(pop_snapshot, by = "Continent_name") %>%
  arrange(desc(mean_hpi)) %>%
  mutate(
    xmax = cumsum(total_pop),
    xmin = lag(xmax, default = 0),
    Continent_name = factor(Continent_name, levels = Continent_name)
  )

# ---- Plot ----
p1 <- ggplot(chart1_data) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = mean_hpi,
    fill = Continent_name),
    color = "white") +
  scale_fill_viridis_d(name = NULL) +
  scale_x_continuous(labels = label_number(scale = 1e-3, suffix = "M")) +
  labs(title = "Latin America achieves the highest wellbeing despite a smaller
population",
    x = "Cumulative Population (millions)", y = "Mean Happy Planet Index",
    caption = "Source: Happy Planet Index, 2006-2025 public dataset") +

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theme_hpi_team()

print(p1)
print(chart1_data)

ggsave("Chart1_HPI_by_Continent.jpg",
       plot = p1, width = 9, height = 6, dpi = 300, bg = "white")

# =====
# Chart 2: Table with embedded charts
# =====

# PACKAGES -----
library(dplyr)
library(readxl)
library(gt)
library(gtExtras)
library(webshot2)
library(svglite)

# ---- CONFIG -----
file_path <- r"(F:\UTDallas_class\Data Visualization\Assignment 4 files\Happy-
Planet-Index-2006-2025-public-data-set (2).xlsx)"

cats <- tribble(
  ~id,      ~col,                                     ~label,
  ~color,   ~dec,                                     "HPI",
  "cat1", "HPI",                                     "HPI",
  "#6a4c93", 1,                                     "Life Satisfaction",
  "cat2", "Life Satisfaction",                       "Life Satisfaction",
  "(0-10)",  "#2a9d8f", 1,                           "Life Expectancy",
  "cat3", "Life Expectancy",                         "Life Expectancy",
  "(yrs)",  "#457b9d", 1,                           "Ecological",
  "cat4", "Ecological Footprint",                     "Ecological",
  "Footprint (gha)", "#e76f51", 1,                   "GDP per capita",
  "cat5", "GDP per capita (cleaned) - 2024 used for 2025", "GDP per capita",
  "(USD)",  "#e9c46a", 0
)

spark_pal <- c("grey30", "grey30", "red", "darkgreen", "lightgrey")

# ---- 1. READ -----

# Read the HPI Excel file
hpi <- read_excel(
  file_path,
  sheet = "All Data",
  skip = 1
) %>%
  rename(HPI_rank = `HPI rank`) %>%
  select(
    Country,
    Year,
    HPI_rank,
    all_of(setNames(cats$col, cats$id))
  ) %>%
  mutate(
    across(c(HPI_rank, all_of(cats$id)), as.numeric)
  )

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# ---- 2. 2025 RANKED COUNTRIES -----
```

```
hpi_2025 <- hpi %>%  
  filter(  
    Year == 2025,  
    !is.na(HPI_rank)  
  ) %>%  
  select(  
    Country,  
    HPI_rank  
  )
```

```
# ---- 3. LATEST VALUE + TREND PER COUNTRY -----
```

```
trend_tbl <- hpi %>%  
  filter(  
    Country %in% hpi_2025$Country,  
    Year <= 2025  
  ) %>%  
  arrange(  
    Country,  
    Year  
  ) %>%  
  group_by(Country) %>%  
  summarise(  
    across(  
      all_of(cats$id),  
      list(  
        latest = ~ last(na.omit(.x)),  
        trend = ~ list(as.numeric(na.omit(.x)))  
      ),  
      .names = "{.col}_{.fn}"  
    ),  
    .groups = "drop"  
  ) %>%  
  filter(  
    if_all(  
      ends_with("_trend"),  
      ~ lengths(.x) >= 2  
    )  
  )
```

```
# ---- 4. FIRST 15 COUNTRIES -----
```

```
tbl_data <- hpi_2025 %>%  
  inner_join(  
    trend_tbl,  
    by = "Country"  
  ) %>%  
  arrange(HPI_rank) %>%  
  slice_head(n = 15)
```

```
# ---- 5. COLUMN ORDER -----
```

```
col_order <- c(  
  "Country",  
  "HPI_rank",  
  unlist(  
    lapply(  
      cats$id,
```

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      \ (i) paste0(i, c("_latest", "_trend"))
    )
  )
)

# ---- 6. TABLE -----

use_sparklines <- TRUE

# Function to create inline bars
make_bar <- function(max_val, color, dec) {

  force(max_val)
  force(color)
  force(dec)

  function(x) {

    v <- suppressWarnings(as.numeric(x))

    w <- round(v / max_val * 70)

    w[is.na(w)] <- 0

    num <- ifelse(
      is.na(v),
      "",
      formatC(
        v,
        format = "f",
        digits = dec,
        big.mark = ",",
      )
    )

    paste0(
      '<div style="display:flex;align-items:center;gap:6px;">',
      '<div style="background:', color,
      ';height:12px;width:', w,
      'px;border-radius:2px;"></div>',
      '<span style="font-size:11px;">',
      num,
      '</span>',
      '</div>'
    )
  }
}

# Create gt table
tbl <- tbl_data %>%
  select(all_of(col_order)) %>%
  gt()

# ---- 7. ADD BARS AND SPARKLINES -----

for (k in seq_len(nrow(cats))) {

  id      <- cats$id[k]
  lat_col <- paste0(id, "_latest")
  trd_col <- paste0(id, "_trend")

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mx <- max(
  tbl_data[[lat_col]],
  na.rm = TRUE
)

# Add inline bar
tbl <- tbl %>%
  text_transform(
    locations = cells_body(
      columns = all_of(lat_col)
    ),
    fn = make_bar(
      mx,
      cats$color[k],
      cats$dec[k]
    )
  )

# Add sparkline
if (use_sparklines) {
  tbl <- tbl %>%
    gt_plt_sparkline(
      !!rlang::sym(trd_col),
      type = "shaded",
      label = FALSE,
      palette = spark_pal
    )
}

# Add indicator heading
tbl <- tbl %>%
  tab_spanner(
    label = cats$label[k],
    columns = starts_with(id)
  )

# Add subheadings
tbl <- tbl %>%
  cols_label(
    !!rlang::sym(lat_col) := "2025 Performance",
    !!rlang::sym(trd_col) := "Trend 2006-25"
  )
}

# ---- 8. COLUMN LABELS -----

tbl <- tbl %>%
  cols_label(
    Country = "Country",
    HPI_rank = "HPI Rank"
  )

# ---- 9. SOURCE -----

tbl <- tbl %>%
  tab_source_note(
    source_note = html(
      '<div style="text-align:right; font-size:9px;">
        Source: Happy Planet Index 2006-2025 Public Dataset
      </div>'
    )
  )

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# ---- 10. DISPLAY TABLE -----
print(tbl)

# ---- 11. SAVE AS PNG -----

gtsave(
  tbl,
  r"(F:\UTDallas_class\Data Visualization\Assignment 4
files\hpi_table_first15.png)"
)

# -----
# Chart 3:Bar chart
# Chart 4:Column chart
# -----
# -----
# 1. PACKAGES
# -----

library(readxl)
library(dplyr)
library(ggplot2)
library(forcats)

# -----
# 2. LOAD THE HPI DATA
# -----

file_path <- "C:/Users/mdpav/Downloads/Happy-Planet-Index-2006-2025-public-data-
set.xlsx"

hpi_data <- read_excel(
  file_path,
  sheet = "All Data",
  skip = 1
)

# Check column names
print(names(hpi_data))

# -----
# 3. SELECT 2025 DATA
# -----

hpi_2025 <- hpi_data %>%
  filter(
    Year == 2025,
    !is.na(HPI)
  )

# Check number of observations
print(nrow(hpi_2025))

# =====
# CHART 3

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# TOP AND BOTTOM 20 COUNTRIES BY HPI
# =====

# -----
# 4. TOP 20 COUNTRIES
# -----

top20 <- hpi_2025 %>%
  slice_max(
    order_by = HPI,
    n = 20,
    with_ties = FALSE
  ) %>%
  mutate(
    group = "Top 20"
  )

# -----
# 5. BOTTOM 20 COUNTRIES
# -----

bottom20 <- hpi_2025 %>%
  slice_min(
    order_by = HPI,
    n = 20,
    with_ties = FALSE
  ) %>%
  mutate(
    group = "Bottom 20"
  )

# -----
# 6. COMBINE TOP AND BOTTOM 20
# -----

chart3_data <- bind_rows(
  bottom20,
  top20
)

# Check the selected countries
print(
  chart3_data %>%
    select(Country, HPI, group) %>%
    arrange(HPI)
)

# -----
# 7. COLOUR-BLIND-SAFE PALETTE
# -----

hpi_colors <- c(
  "Top 20" = "#0072B2",
  "Bottom 20" = "#D55E00"
)

# -----
# 8. COMMON THEME

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# -----

hpi_theme <- theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(
      face = "bold",
      size = 15
    ),
    plot.subtitle = element_text(
      size = 11
    ),
    axis.title = element_text(
      face = "bold"
    ),
    panel.grid.minor = element_blank(),
    legend.position = "none",
    plot.caption = element_text(
      size = 9,
      hjust = 0
    )
  )

# -----
# 9. CREATE CHART 3
# -----

chart3 <- ggplot(
  chart3_data,
  aes(
    x = fct_reorder(Country, HPI),
    y = HPI,
    fill = group
  )
) +
  geom_col(
    width = 0.7
  ) +
  coord_flip() +
  scale_fill_manual(
    values = hpi_colors
  ) +
  labs(
    title = "Top and bottom 20 countries by Happy Planet Index",
    subtitle = "2025 HPI scores",
    x = "Country",
    y = "Happy Planet Index (HPI)",
    caption = "Source: Happy Planet Index, 2006-2025 public dataset."
  ) +
  hpi_theme

# Display Chart 3

print(chart3)

# -----
# 10. SAVE CHART 3
# -----

ggsave(
  filename = "chart3_top_bottom_20_hpi.png",
  plot = chart3,

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```

width = 10,
height = 8,
dpi = 300
)

# =====
# CHART 4
# MEAN LIFE SATISFACTION BY CONTINENT
# =====

# -----
# 11. CREATE CONTINENT LABELS
# -----

hpi_2025 <- hpi_2025 %>%
  mutate(
    continent_name = case_when(
      Continent == 1 ~ "Latin America",
      Continent == 2 ~ "N America & Oceania",
      Continent == 3 ~ "Western Europe",
      Continent == 4 ~ "Middle East & N Africa",
      Continent == 5 ~ "Africa",
      Continent == 6 ~ "South Asia",
      Continent == 7 ~ "Eastern Europe & Central Asia",
      Continent == 8 ~ "East & Southeast Asia",
      TRUE ~ NA_character_
    )
  )

# -----
# 12. CALCULATE MEAN LIFE SATISFACTION
# -----

chart4_data <- hpi_2025 %>%
  filter(
    !is.na(`Life Satisfaction`),
    !is.na(continent_name)
  ) %>%
  group_by(continent_name) %>%
  summarise(
    mean_life_satisfaction = mean(
      `Life Satisfaction`,
      na.rm = TRUE
    ),
    .groups = "drop"
  )

# Check calculated means

print(chart4_data)

# -----
# 13. CREATE CHART 4
# -----

chart4 <- ggplot(
  chart4_data,
  aes(
    x = fct_reorder(

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```

        continent_name,
        mean_life_satisfaction
    ),
    y = mean_life_satisfaction
)
) +
geom_col(
    fill = "#0072B2",
    width = 0.7
) +
labs(
    title = "Mean life satisfaction by continental group",
    subtitle = "2025",
    x = "Continental group",
    y = "Mean life satisfaction",
    caption = "Source: Happy Planet Index, 2006–2025 public dataset."
) +
hpi_theme +
theme(
    axis.text.x = element_text(
        angle = 30,
        hjust = 1
    )
)
)

```

Display Chart 4

```
print(chart4)
```

```

# -----
# 14. SAVE CHART 4
# -----

```

```

ggsave(
    filename = "chart4_mean_life_satisfaction_continent.png",
    plot = chart4,
    width = 10,
    height = 6,
    dpi = 300
)

```

```

# =====
# 15. SESSION INFORMATION
# =====

```

```
sessionInfo()
```